# SOWMYA NAGULA

Software Engineer

4693562259 | nagulasowmya5@gmail.com | United States | LinkedIn

## SUMMARY

Software Engineer with 3+ years of experience designing and deploying scalable distributed systems, AI-enabled applications, and full-stack platforms in production environments. Proven expertise in JavaScript/TypeScript, Java, Python, and Go with strong foundations in data structures, algorithms, and large-scale system design. Experienced in building API-driven microservices, cloud-native infrastructure on GCP/AWS, and production-grade AI/ML workflows, including Generative AI and intelligent agent development for enterprise cloud solutions.

## EDUCATION

**Master's in Computer Science**, University of North Texas — 2022 — 2024, Denton, TX, United States

**Bachelor's in Computer Science**, Sreyas Institute of Engineering and Technology — 2017 — 2021, HYDERABAD

## SKILLS

**Programming Languages:** C, C++, Java, Python, JavaScript, TypeScript, SQL, Go, Kotlin

**AI / ML & Generative AI:** Large Language Models (LLMs), Prompt Engineering, Model Evaluation, NLP, Conversational AI systems, ML pipelines, AI system monitoring

**Frontend:** React, TypeScript, Component Architecture, Accessibility (WCAG), State Management, Client-side Performance Optimization

**Backend & Distributed Systems:** REST APIs, Microservices, Distributed Systems, High-Availability Services, Concurrency, System Design

**Cloud & Infrastructure:** Google Cloud Platform (GCP), AWS, Azure, Kubernetes, Docker, Terraform, CI/CD, Cloud Storage, Compute services

**Testing & Reliability:** JUnit, Pytest, Integration Testing, Performance Testing, Automated Testing, SLA-driven incident response, Debugging production systems

## WORK EXPERIENCE

**Wells Fargo** — Feb '24 — Present
**Software Engineer** — Dallas, TX, United States

- Designed and deployed scalable microservices using Java, Python, and Spring Boot supporting distributed financial systems handling 1M+ daily transactions.
- Built and maintained RESTful APIs consumed by internal AI-powered analytics platforms, improving response latency by 28%.
- Contributed to distributed infrastructure reliability by implementing observability (logging, monitoring, tracing), reducing MTTR by 35%.
- Led design reviews across cross-functional teams (8+ engineers), influencing architectural decisions for high-availability systems.
- Improved system scalability by optimizing database indexing and caching strategies, reducing service response time by 40%.
- Supported production systems under strict SLAs, troubleshooting networking and compute-related incidents to minimize customer impact.
- Contributed to internal intelligent automation workflows leveraging NLP-based classification models for document processing.

**Zaven IT Solutions** — May '23 — Jan '24
**Software Engineer** — Irving, TX, United States

- Developed full-stack applications using React, TypeScript, Node.js, and PostgreSQL serving 50K+ monthly users.
- Built API-driven backend services enabling real-time data synchronization across distributed systems.
- Designed modular UI components with accessibility standards compliance, improving usability scores by 22%.

- Implemented CI/CD pipelines and automated testing frameworks increasing deployment reliability by 30%.
- Collaborated in Agile teams and participated in code reviews to enforce code quality, scalability, and efficiency standards.
- Optimized client-side rendering and reduced page load time by 35% through performance profiling and code splitting.

**Accenture** May '21 — Dec '21
**Software Engineer** Hyderabad, India

- Engineered enterprise backend systems using Java and Spring deployed on AWS and Azure cloud infrastructure.
- Designed and optimized REST APIs integrated with distributed storage systems.
- Resolved scalability bottlenecks in high-volume data processing workflows, improving throughput by 25%.
- Participated in system debugging, performance profiling, and root cause analysis of production issues.

## ACADEMIC PROJECTS

**iCARE – Intelligent Clinical Assignment and Record Entry System** Aug '22 — Dec '22

- Developed and executed automated test cases using Selenium WebDriver with Java/Python, validating pen-based login, patient assignment, and document creation workflows.
- Created UI test scripts simulating interactions for admin, doctor, and nurse roles, ensuring correct dynamic content rendering, form validation, and role-based access control using XPath, TestNG, and POM (Page Object Model).
- Integrated Selenium testing into CI/CD pipelines with JUnit/TestNG and GitHub, improving regression coverage and system reliability across browsers and screen resolutions.

**A comprehensive study on SNMDs via online social media mining** Jul '20 — Oct '20

- Proposed a machine learning framework leveraging social network features to detect potential SNMD cases.
- Developed a new SNMD-based Tensor Model (STM) to enhance detection accuracy.
- Evaluated framework with a user study of 3,126 online social network participants, demonstrating improved predictive performance.

## PUBLICATIONS

**Instant Augmented Reality Converter Tool**
Scopus International Journal Publication
No-133 https://gisscience.net/VOLUME-8-ISSUE-6-2021/
Paper ID: GSJ4300
• Developed a web application enabling users to upload 3D models via an Upload Tool. Applied textures/skins to 3D models in a single click and converted 3D models into Android (GLTF-Web) and iOS (USDZ-Apple) formats. Implemented a single-click feature merging digital and physical worlds, delivering an immersive, interactive user experience.
•Tool used JavaScript, Node Js, OBJ Models, Local Server hosting, ARCore, ARKit, Depth sensing camera.